

Smart Outlet System

Chris James, V0096666029 | Colette Reimer, V00929031 | Olivia Moonie, V00952653

Introduction

This project presents a low-cost smart outlet system that enables remote power control and energy monitoring through a central hub and Powerline Communication (PLC). Designed with a focus on privacy and accessibility the prototype provides a foundation for practical smart home automation.

Design Goals

The purpose of this project was to build a proof-concept device for a smart outlet system that

- allows the user to control their outlets remotely,
- reports power consumption data to the user,
- respects the user's privacy, and
- can be progressively rolled out in the user's home without regard for wireless connectivity.

An encrypted connection from the user's device to the hub over the internet allows the user to toggle the outlet and receive power consumption data from anywhere in the world. User data is never stored or accessible on any device outside their control. Communication from the hub to the outlet is facilitated by power-line communication (PLC). This provides connectivity throughout the house or property without regard for wireless meshing [1].

Existing solutions rely on wireless connectivity between the outlet and hub, which can face problems in large homes or where smart outlets are sparsely used [2], [3]. Data is also typically stored in the cloud, rather than on the hub or user's device [2], [3]. In an age of increasing surveillance and eroding privacy, many users want a solution that allows them to retain control of their data.

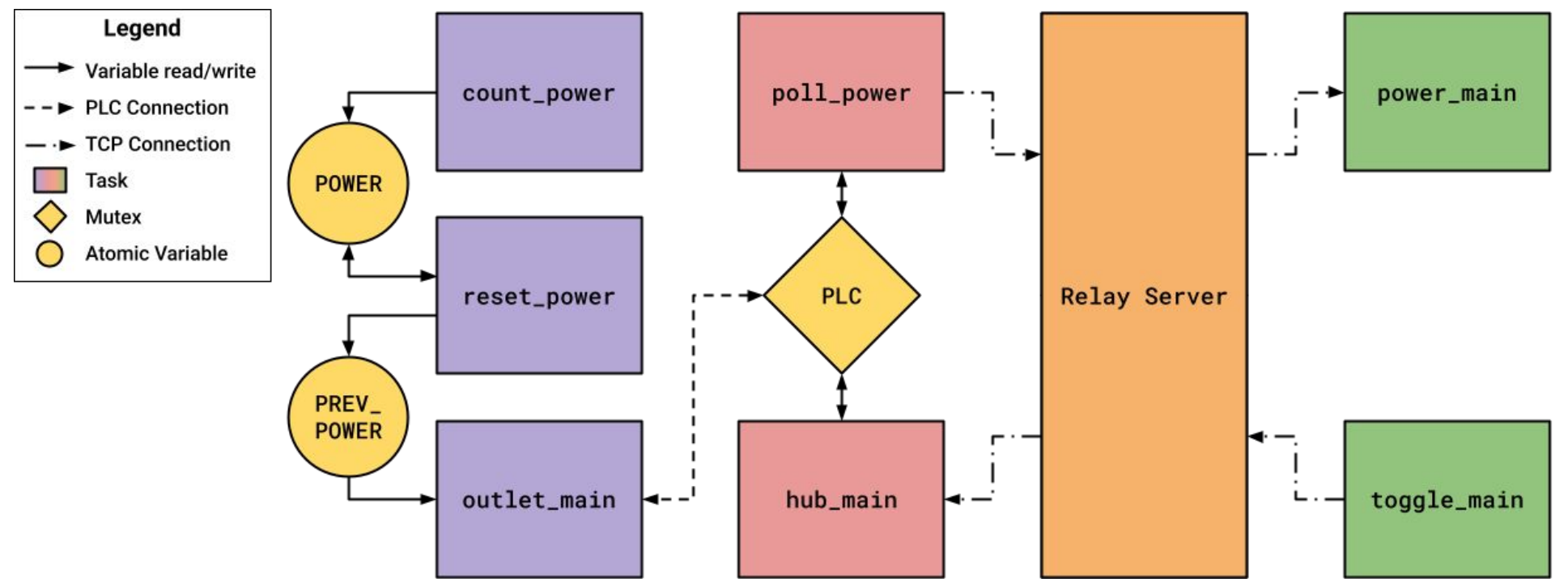


Figure 1: Software Overview Diagram

Hardware Architecture

Process for PCB Design

- Design was made in KiCAD
- Symbols and footprints via Digikey, Mouser, Ultra Librarian and SnapMagic
- Datasheets used for pin connections
- Equation 1 used for calculating trace widths
- Extra layers were added to negate the need for extreme sized traces
- Most SMD attached using reflow oven
- Through-hole and some SMD attached via hand held soldering tool

$$EQN\ 1: W = \left(\frac{I}{kT^{0.5} Rise} \right)^{\frac{1}{\epsilon}} [4]$$

Component Selection Process

- Review of components selected for ECE 399 Project (Major Components only)
- Prioritized SMD when possible to reduce area required
- Manufacturers of the PLC module and power measurement component provided suggested circuits ([5] & [6], respectively)
- IRM-05-5 was selected to convert the high voltage, high current AC to low voltage, low current DC [7]
- Power requirements of each branch to confirm ratings were sufficient using equation 2

$$EQN\ 2: P = \frac{V^2}{R} = VI = I^2R$$

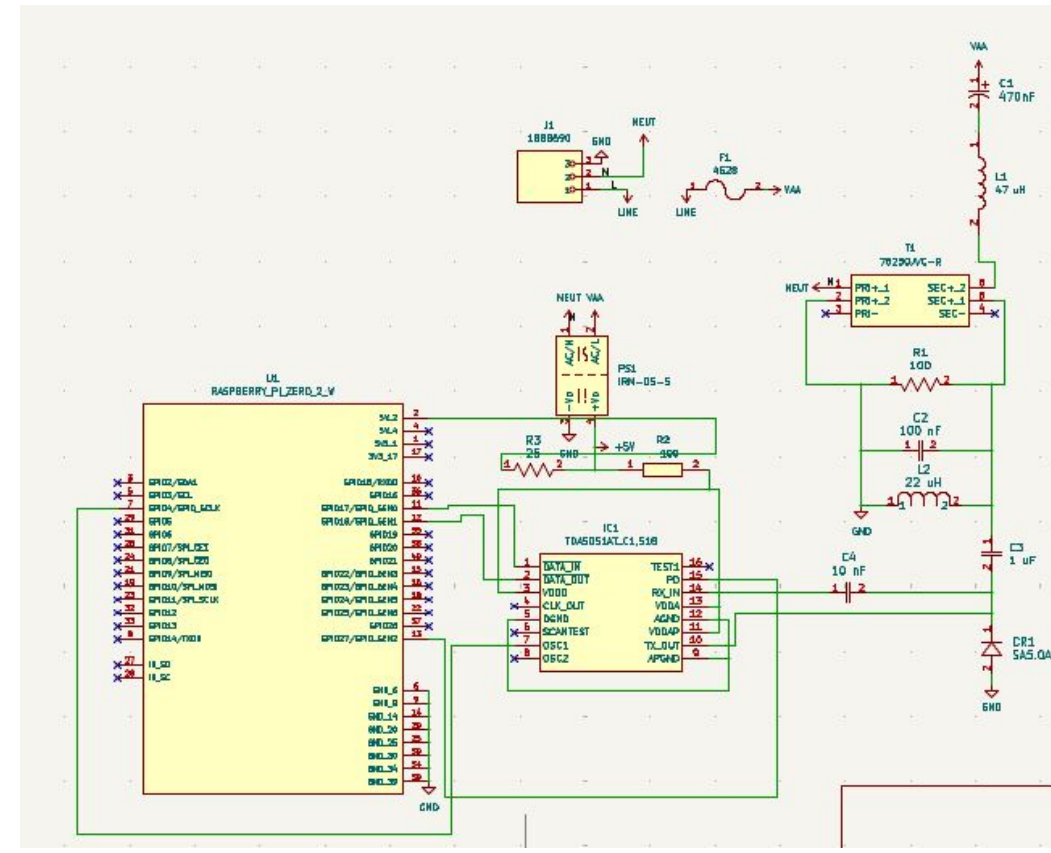


Figure 3: Hub Connections

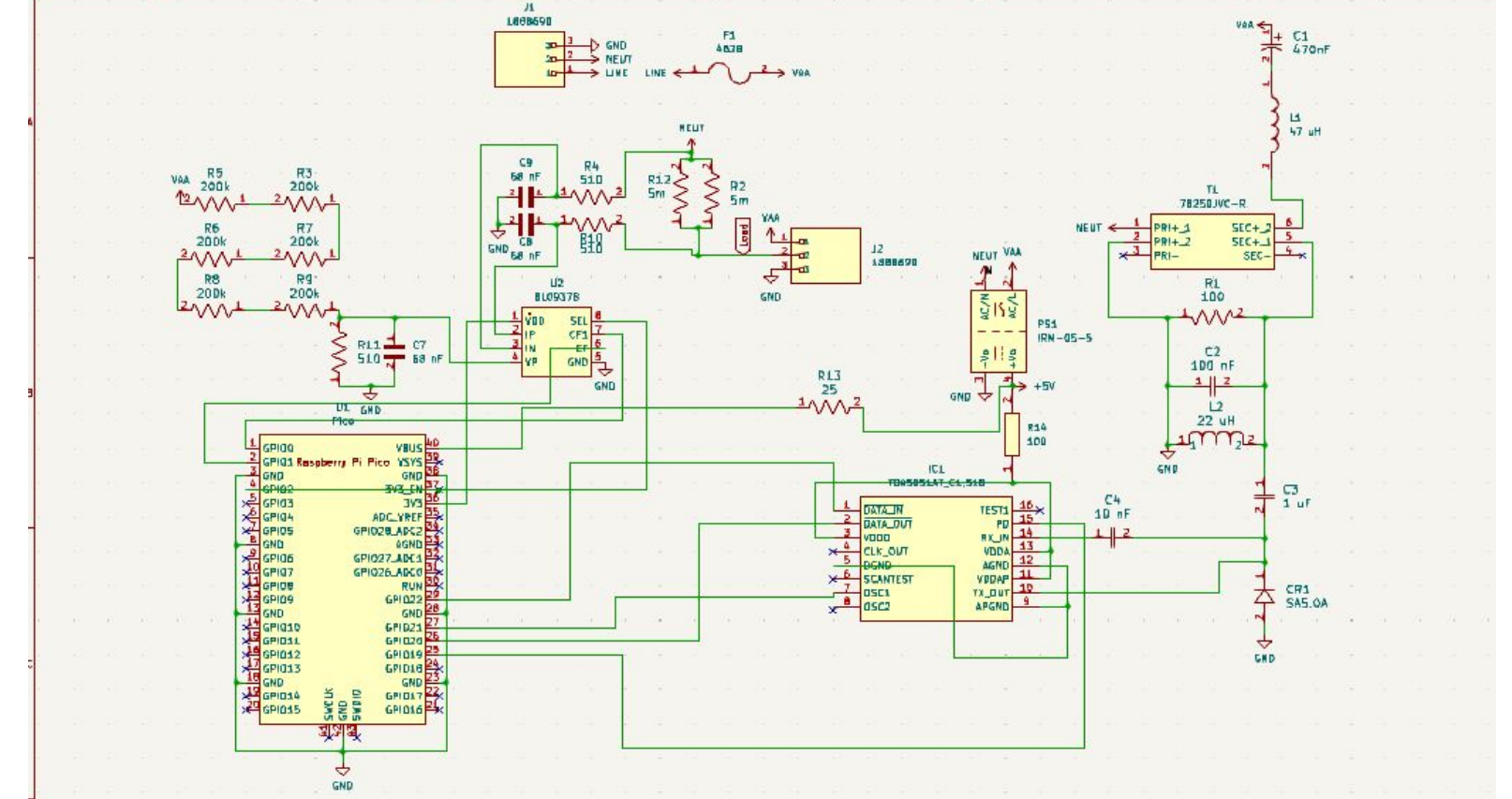


Figure 4: Outlet Connections

Software Architecture

There are four different platforms present in the smart outlet system. Figure 1 shows an overview.

- The outlet, using a bare-metal Raspberry Pi Pico 2
- The hub, using a Raspberry Pi Zero 2W running Debian 13
- The user device, running any mainstream OS
- The relay server, running Debian 13

Software for the outlet, hub, and user device were written in Rust. Software for the outlet and hub was written as a collection of concurrent tasks using the Embassy async runtime for embedded Rust, similar to programming for an RTOS. Communication between the hub and user device was facilitated by Wireguard, a fast and secure VPN included with the Linux kernel [8]. The relay server allows the hub and user device to communicate across multiple NATs (network address translation). Systemd services were used to coordinate the startup processes of the hub, and Iptables was used with kernel-level forwarding to enable the relay server to forward Wireguard traffic between the hub and user device. Figures 6 and 7 show flow charts for the hub and and outlet programs, respectively.

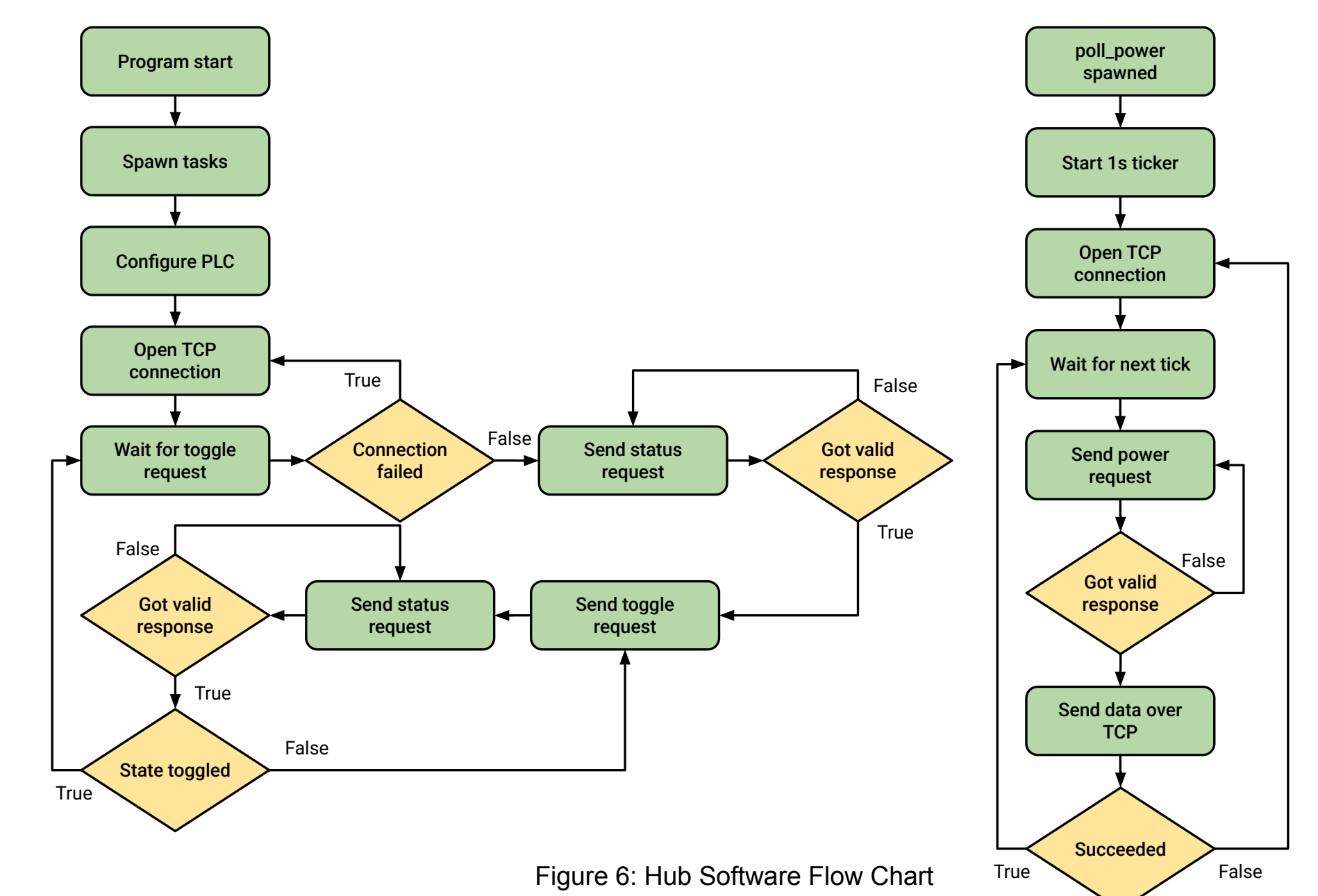


Figure 6: Hub Software Flow Chart

Conclusion and Future Considerations

The project successfully demonstrates a functional smart outlet prototype with remote control and energy monitoring capabilities.

Future Considerations

- Integrate a functional AC receptacle for direct wall installation
- Redesign the electronics using a smaller, more compact PCB
- Improve hardware reliability and electrical safety
- Develop a user-friendly smartphone app for control and data aggregation
- Expand energy monitoring and automation features

References and Acknowledgements

[1] P. Goyal, "Design of Power-Line Communication System (PLC) Using a PIC Microcontroller," International Conference on Information & Communication Technology (IICT), DIT University, India, July 2007

[2] C. Lien, Y. Bai, and M. Lin, "Remote-Controllable Power Outlet System for Home Power Management", IEEE Transactions on Consumer Electronics, vol.53, no. 4, pp. 1634-1641, Nov. 2007, doi: 10.1109/TCE.2007.4429263.

[3] A.S. Musleh, M. Debouza, M. Farook, "Design and implementation of smart plug: An Internet of Things (IoT) approach," IEEE, Jan. 2018, doi: 10.1109/ICCTA.2017.8252033

[4] "PCB Trace Width Calculator." Digikey. <https://www.digikey.ca/en/resources/conversion-calculators/conversion-calculator-pcb-trace-width> (Accessed July 26, 2026)

Enclosure Design

Process for Enclosure Design

- Designed in Autodesk Fusion 360 using a measurement-based approach
- Initial concept sketch used to determine component layout
- KiCad PCB model imported to accurately position mounting features and cutouts
- Multiple design iterations to optimize fit, size, and assembly
- Dimensions verified using the physical PCB before printing
- Two-piece (lid and base) enclosure designed for both the Hub and Outlet
- Exported as a 3MF file and 3D printed in white PLA using Bambu Studio

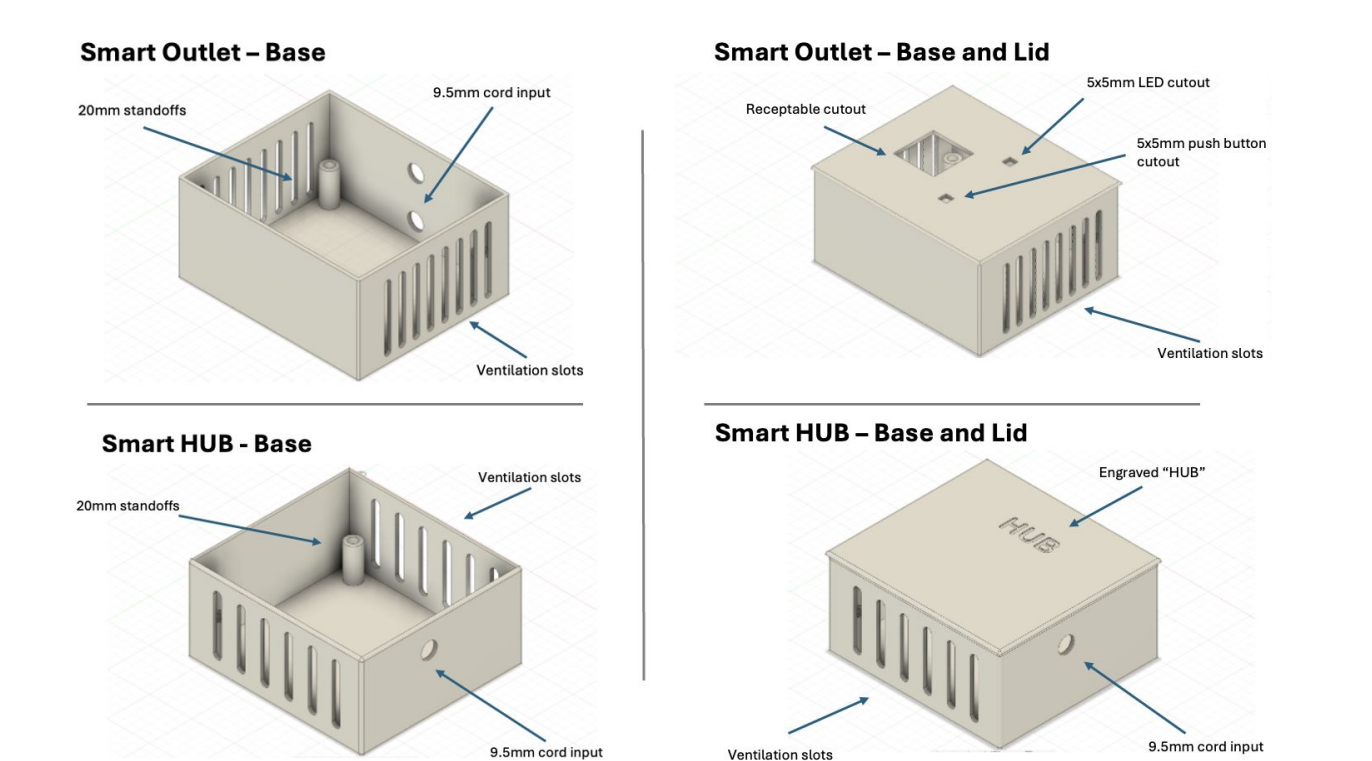


Figure 5: Diagram of Autodesk Fusion 360 Enclosures

[5] Single Phase Energy Meter IC with Integrated Oscillator for Socket, Shanghai Belling, Accessed: July 26, 2026. [Online]. Available: https://www.belling.com.cn/media/file_object/bel_product/BL0937B/datasheet/BL0937B_V1.0_en.pdf

[6] TDA5051A Home automation modem, Rev. 5 (2011). NXP Semiconductors, Accessed: July 26, 2026. [Online]. Available: <https://www.mouser.ca/datasheet/3/118/1/TDA5051A.pdf>

[7] 5W AC-DC High Reliable PCB-Mount Green Industrial Power Module, Mean Well, Accessed: July 26, 2026. [Online]. Available: <https://www.meanwellusa.com/upload/pdf/IRM-05/IRM-05-spec.pdf>

[8] S. Zakhary, T. Lodge, D. McAuley. "Performance Evaluation for Privacy-preserving Control of Domestic IoT Devices." 2022. [Online]. Available: <https://arxiv.org/abs/2207.08482>

Special thanks to Rob Fichtner, for his advice and the help he provided.

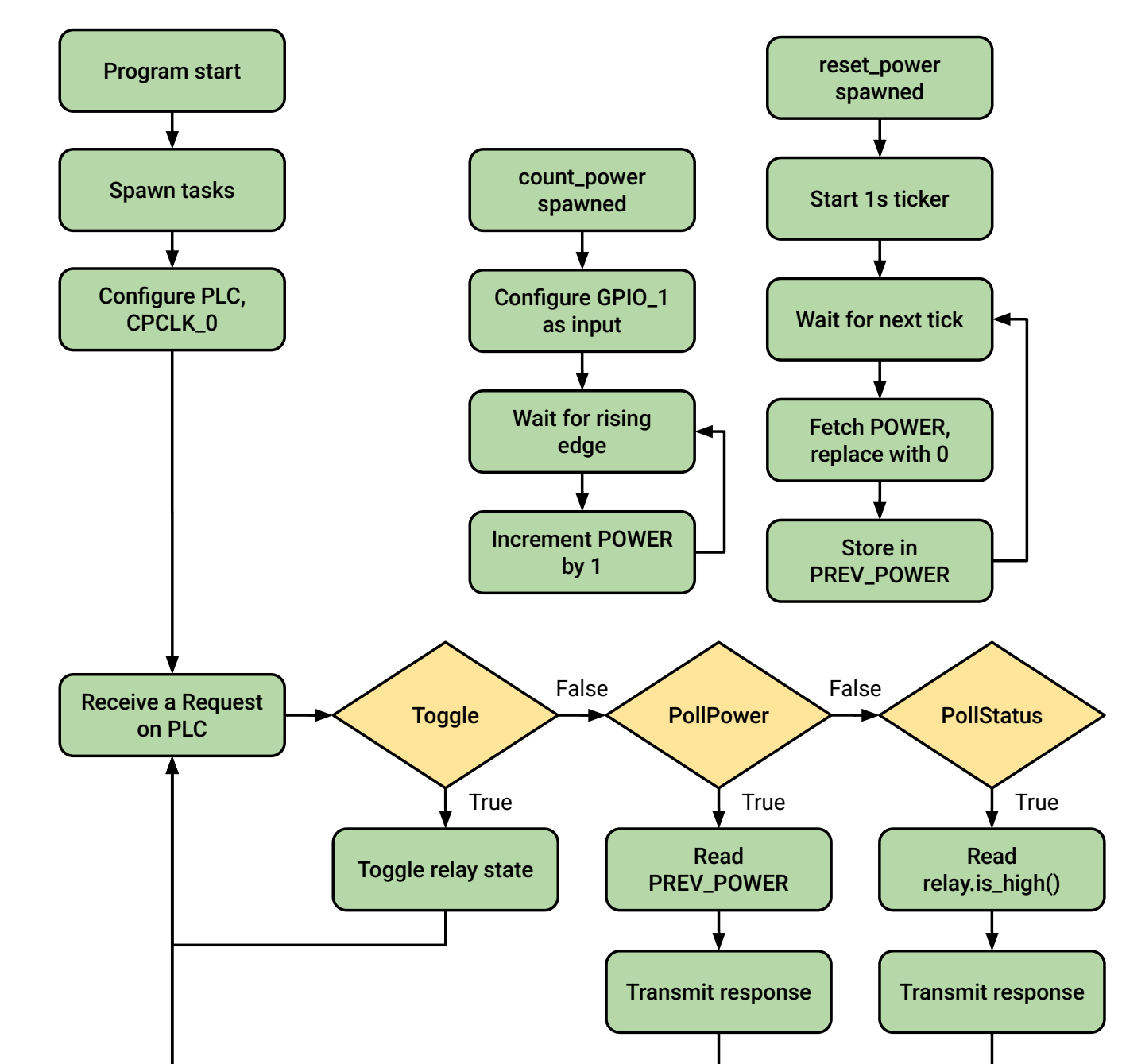


Figure 7: Outlet Software Flow Chart

The outlet and hub communication over PLC using a fixed-length half-duplex protocol (meaning every device shares a single line). The hub periodically sends requests to the outlet to send power data, send the outlet's state, or toggle the outlet's state. The hub will resend requests for data until it receives a valid response from the outlet. When toggling the outlet, the hub first gets the current state, then sends a toggle request, then gets the state again; this process repeats until the outlet's state has successfully toggled. The outlet never initiates a conversation.